Pop rivets are a quick and easy alternative that was developed within the aircraft industry to enable riveting from just one side. Pop rivets are hollow and therefore not as strong as ordinary rivets.

MACHINE SCREWS, NUTS AND BOLTS

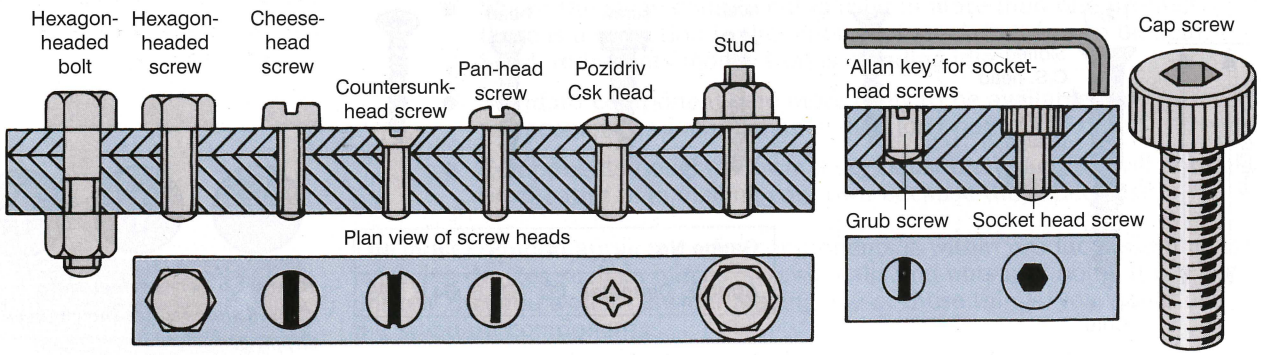


Figure 8.10 *Machine screws and bolts.*

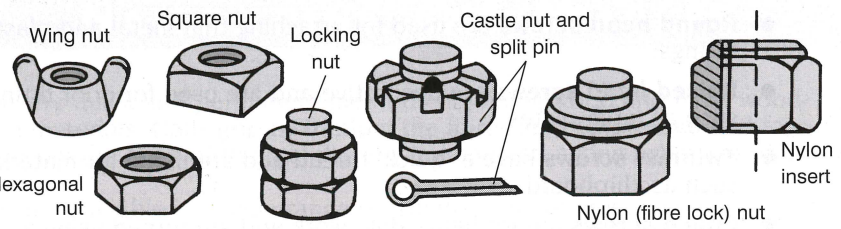


Figure 8.11 *Types of nut.*

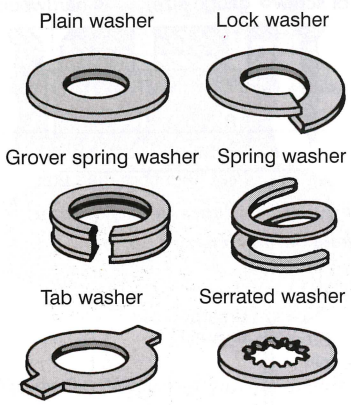


Figure 8.12 *Types of washer.*

Machine screws, nuts and bolts are regarded as temporary fittings because they can easily be taken apart.

- **Machine screws** are available in a wide range of materials, diameters, lengths, head shapes and thread forms.
- **Bolts** are strong being made from high tensile steel. They have either square or hexagonal heads.
- **Nuts** must match, in size and thread form, the screw or bolt with which they are to be used. There are many forms of nut, from those that are designed to be easily removed (wing nuts) to those that resist removal, particularly through vibration (lock nuts).
- **Washers** are used to protect the surface when nuts are being tightened by spreading the load. They also help to prevent loosening through vibration.

Thread forms

The ISO standard thread form is metric-sized ISO coarse and ISO fine. There are however many other thread forms still in regular usage. In particular you may come across BA series threads and also BSF, BSW, UNC and UNF.

Self-tapping screws

Self-tapping screws are made from hardened steel and cut their own thread as they are screwed in. They are suitable for joining sheet metal and plastics. Preparation requires a clearance hole and a pilot hole equal to the screw's core diameter.

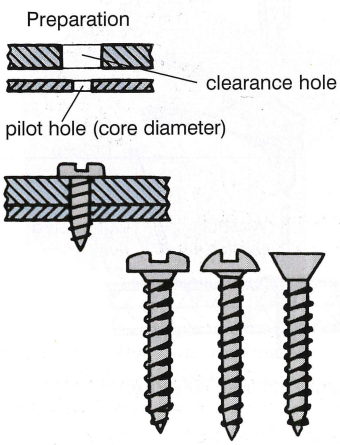


Figure 8.13 *Self-tapping screws.*